



Temperature, Heat and 1st Law of Thermodynamics

Temperature 0th Law of Thermodynamics
 Thermal expansion Heat and Work
 1st Law of Thermodynamics

Temperature

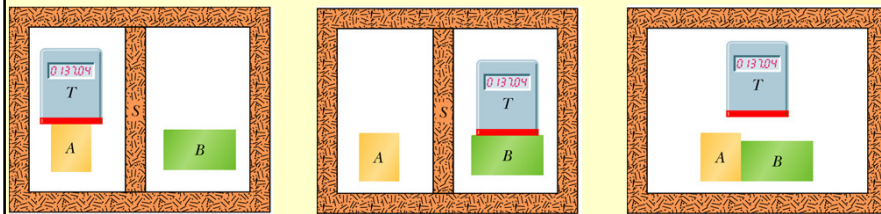
- SI unit: Kelvin – K
- Other units: $T_C = (T_K - 273.15)$

$$T_F = \frac{9}{5}T_C + 32^\circ$$

- C° - change in T

0th Law of Thermodynamics

- Properties of many materials change with temperature:
 - Liquid's volume $\uparrow$ when temp. $\uparrow$
 - Metal rod increases in length with increase in temp.
 - Electrical resistance of a wire increases with increase in temp.
 - Pressure of enclosed gas increases with increase in temp.
- **0th Law of Thermodynamics:** If body A and body B are each in thermal equilibrium with a third body T, then A and B are in equilibrium with each other.



3

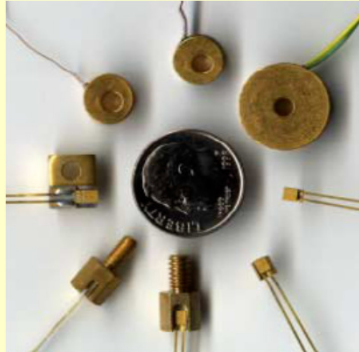
Measuring Temperature

- Triple point of water
 - Standard fixed point with a fixed temperature
 - $T_3 = 273.16$ K (not °K)

4

Thermal properties

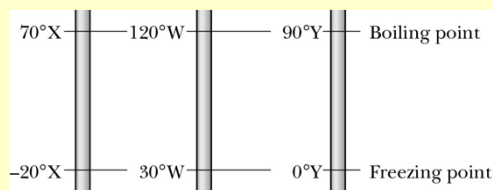
- Thermocouples



5

Checkpoint 1

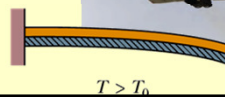
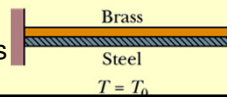
- The figure shows 3 linear temperature scales with freezing and boiling points of water indicated. (a) Rank the degrees on these scales by size. (b) Rank the following temperatures highest first: 50°X , 50°W and 50°Y .



6

Thermal Expansion

- What happens when the temperature of a material increases?
 - Energy is added
 - Atoms move a bit further apart
 - Different materials' atoms move different distances further apart
- Thermal expansion must be taken into account in everyday situations or used to our advantage.
 - Bridge, train tracks and buildings have expansion joints
 - Dental fillings
 - Concorde (due to frictional forces)
 - Bimetal strip
 - Thermostats
 - Thermometers



7

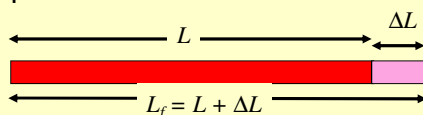
Thermal Expansion



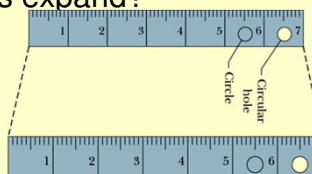
8

Linear Expansion

- Consider a metal rod of length L which is raised by a temperature ΔT .



- Its **length will increase** from L to L_f : $\Delta L = L\alpha\Delta T$
- α is the **coefficient of linear expansion**
- Units of α : $^{\circ}\text{C}^{-1}$ or K^{-1}
- How do holes and voids in objects expand?
 - Holes
 - Circle
 - Scale
 - Numbers



9

Linear Expansion

Table 18-2

Some Coefficients of Linear Expansion^a

Substance	α ($10^{-6}/^{\circ}\text{C}^{\circ}$)	Substance	α ($10^{-6}/^{\circ}\text{C}^{\circ}$)
Ice (at 0°C)	51	Steel	11
Lead	29	Glass (ordinary)	9
Aluminum	23	Glass (Pyrex)	3.2
Brass	19	Diamond	1.2
Copper	17	Invar ^b	0.7
Concrete	12	Fused quartz	0.5

10

Volume Expansion

- All dimensions increase with an increase in temperature.
- Volume must also increase.
- $\Delta V = V \beta \Delta T$
- β is the **coefficient of volume expansion**
- $\beta = 3\alpha$

11

Why does ice float in water?

- Above 4 °C water expands with an increase in temperature like other solids – density decreases.
- Between 0 to 4 °C water contracts with increase in temperature.
- At 4 °C water has a max. density.

12

Problem

- A steel train track has a length of 30 m when the temperature is 32 °F. What is its length when the temperature is (a) 313 K? (b) -40 °C?

13

Sample Problem p483

- On a hot day in LA, an oil trucker loaded 37 000 L of diesel fuel. He encountered cold weather on his journey, and where it was 23.0 K lower than in LA, he delivers his entire load. How many litres did he deliver? The coefficient of volume expansion for diesel is $9.50 \times 10^{-4}/\text{C}^\circ$, and the coefficient of linear expansion for the steel tank is $11 \times 10^{-6}/\text{C}^\circ$.

14

Temperature and Heat

- **Thermal energy** is an internal energy that consists of the kinetic and potential energies of the random motion of the atoms and molecules in the object.
- **Heat Q** : is the energy transferred between a system and its environment because of a temperature difference between them.
- Heat is a form of energy – measured in Joule
- Other common units:
 - 1 calorie (cal) = 4.1860 J
 - 1 Calorie (Cal) = 1kcal = 4 186.0 J [Diet]

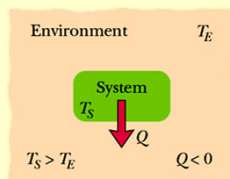


J.P. Joule 1818 – 1889

15

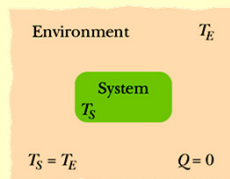
Temperature and Heat

- $T_S > T_E$
- $Q < 0$



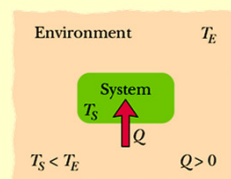
(a)

- $T_S = T_E$
- $Q = 0$



(b)

- $T_S < T_E$
- $Q > 0$



(c)

Sign convention:

Flow out of system: $Q < 0$

Flow into system: $Q > 0$

Internal Energy of a system **CAN** change without energy being transferred by heat. e.g.

- Compress a gas with a piston – no heat flow, but temperature and internal energy increases.
- When a gas expands rapidly – it cools.

16

Absorption of Heat –Solids and liquids

- **Heat capacity C** – proportionality constant (J/K) between Q and ΔT

$$Q = C\Delta T = C(T_f - T_i)$$

- **Specific Heat (Capacity) c** – Heat capacity per unit mass (J/kg·K)

$$Q = cm\Delta T = cm(T_f - T_i)$$

- **Heat of transformation L** – Heat absorbed by the sample when it changes phase – temperature is constant (J kg⁻¹)

$$Q = Lm$$

- L_F heat of fusion (freezing or melting)
- L_V heat of vaporisation (boiling or condensing)

Table 18-3 and 18-4

17

Absorption of Heat

Table 18-3

Some Specific Heats and Molar Specific Heats at Room Temperature

Substance	Specific Heat		Molar Specific Heat
	$\frac{\text{cal}}{\text{g} \cdot \text{K}}$	$\frac{\text{J}}{\text{kg} \cdot \text{K}}$	$\frac{\text{J}}{\text{mol} \cdot \text{K}}$
<i>Elemental Solids</i>			
Lead	0.0305	128	26.5
Tungsten	0.0321	134	24.8
Silver	0.0564	236	25.5
Copper	0.0923	386	24.5
Aluminum	0.215	900	24.4
<i>Other Solids</i>			
Brass	0.092	380	
Granite	0.19	790	
Glass	0.20	840	

Table 18-3

Some Specific Heats and Molar Specific Heats at Room Temperature

Substance	Specific Heat		Molar Specific Heat
	$\frac{\text{cal}}{\text{g} \cdot \text{K}}$	$\frac{\text{J}}{\text{kg} \cdot \text{K}}$	$\frac{\text{J}}{\text{mol} \cdot \text{K}}$
Ice (-10°C)	0.530	2220	
<i>Liquids</i>			
Mercury	0.033	140	
Ethyl alcohol	0.58	2430	
Seawater	0.93	3900	
Water	1.00	4187	

18

Absorption of Heat

Table 18-4

Some Heats of Transformation

Substance	Melting		Boiling	
	Melting Point (K)	Heat of Fusion L_F (kJ/kg)	Boiling Point (K)	Heat of Vaporization L_V (kJ/kg)
Hydrogen	14.0	58.0	20.3	455
Oxygen	54.8	13.9	90.2	213
Mercury	234	11.4	630	296
Water	273	333	373	2256
Lead	601	23.2	2017	858
Silver	1235	105	2323	2336
Copper	1356	207	2868	4730

19

Checkpoint 3

- A certain amount of heat Q will warm 1g of material A by $3\text{ }^\circ\text{C}$ and 1 g of material B by $4\text{ }^\circ\text{C}$. Which material has the greater specific heat?

20

Sample Problem p488

- How much heat must be absorbed by ice of mass $m = 720 \text{ g}$ at -10°C to take it to liquid state at 15°C ?
- 3 step process:
 - Raise temperature from -10°C to 0°C $Q = cm\Delta T = c_i m(T_f - T_i)$
 - Phase change at 0°C – ice to water $Q = L_f m$
 - Raise temperature from 0°C to 15°C $Q = cm\Delta T = c_w m(T_f - T_i)$

21

Sample Problem p488

- If we supply the ice with a total of only 210 kJ (as heat), what are the final temperature and state of the water?:

22

Sample Problem p 487

- A copper slug whose mass m_c is 75 g is heated in a lab oven to 312°C. The slug is then dropped into a glass beaker containing a mass $m_w = 220$ g of water. The heat capacity C_b of the beaker is 45 cal/K. The initial temperature T_i of the water and the beaker is 12 °C. Assuming the slug, beaker and the water are an isolated system and the water does not vaporize, find the final temperature T_f of the system at thermal equilibrium.

$$45 \text{ cal/K} = 188.41 \text{ J/K}$$

	<u>Copper</u>	<u>water</u>	<u>beaker</u>
m (g)	74g	220 g	
C_b (J/K)			188.41
C (J/kg.K)	389	4180	
T_i (°C)	312	12	12
T_f (°C)	?	?	?
	$Q_c = c_c m_c (T_f - T_i)$	$Q_w = c_w m_w (T_f - T_i)$	$Q_b = C_b (T_f - T_i)$

23

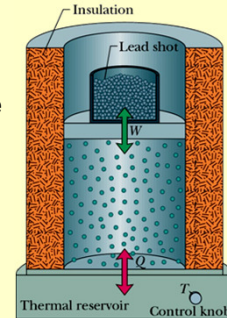
Sample Problem p487

	<u>Copper</u>	<u>water</u>	<u>beaker</u>
m (g)	74g	220 g	
C_b (J/K)			188.41
C (J/kg.K)	389	4180	
T_i (°C)	312	12	12
T_f (°C)	?	?	?
	$Q_c = c_c m_c (T_f - T_i)$	$Q_w = c_w m_w (T_f - T_i)$	$Q_b = C_b (T_f - T_i)$

24

Heat and Work

- Energy can be transferred between system and its environment as heat and work.
 - Gas confined to cylinder with moving piston
 - Upward force on piston due to gas pressure = weight of load on piston.
 - Walls are insulated – no heat transfer
 - Bottom of cylinder is on a thermal reservoir temperature T (controllable)
- Initial State i : p_i, V_i, T_i
- Thermodynamic process results in change to
- Final State f : p_f, V_f, T_f



25

Heat and Work

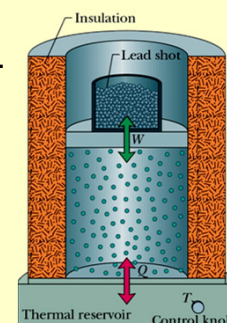
- Thermodynamic process could be:
 - Transfer of energy from reservoir into system: +ve heat
 - Transfer of energy out of the system into reservoir : –ve heat
 - Work can be done by system – raise the piston: +ve work
 - Work can be done on system – lower the piston: -ve work
- **Quasi-static**: Slow change so that the system is always in thermal equilibrium.
- Work done by the system:

$$dW = \vec{F} \cdot d\vec{s}$$

$$= (pA)(ds) \quad p = \frac{F}{A}$$

$$= pdV \quad dV = Ad s$$

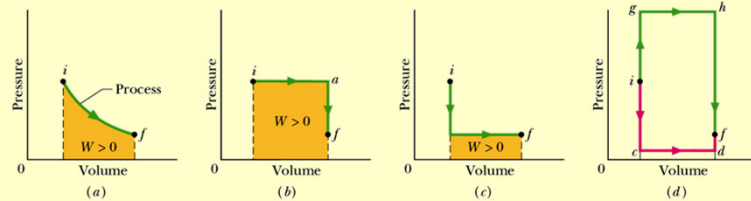
$$W = \int dW = \int_{V_i}^{V_f} p dV$$
 - Area under the pV curve



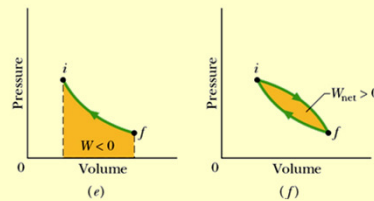
26

Heat and Work

- Many ways for system to get from state i to state f .



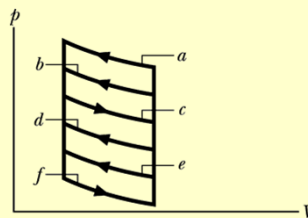
- Sign convention:
 - Gas does work: $W > 0$
 - Work done on gas: $W < 0$
- Changing of state (i to f) by W or Q .
- i to f : $W \neq Q$ and path-dependent



27

Checkpoint 4

- The p - V diagram here shows 6 curved paths (connected by vertical paths) that can be followed by a gas. Which 2 of the curved paths should be part of a closed cycle (those curved paths plus connecting vertical paths) if the net work done by the gas during the cycle is to be at its maximum positive value?



28

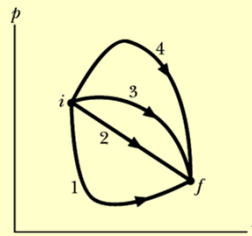
1st Law of Thermodynamics

- When changing state from i to f , Q and W depend on the nature of the process.
- But $Q - W$ is same for all processes – path independent.
- $Q - W = \Delta E_{\text{int}} = E_{\text{int},f} - E_{\text{int},i}$
- **First Law of Thermodynamics**
- $dE_{\text{int}} = dQ - dW$
- The internal energy E_{int} of a system tends to increase if energy is added as heat Q and tends to decrease if energy is lost as work W done by the system.
- Ch 8 – Isolated system – no energy transferred into or out of the system.

29

Checkpoint 5

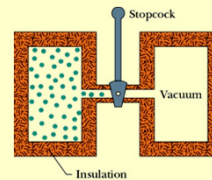
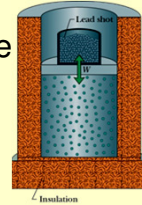
- The figure shows 4 paths on the p - V diagram along which a gas can be taken from state i to f . Rank the paths according to (a) the change ΔE_{int} of the gas, (b) the work done by the gas, and (c) the magnitude of the energy transferred as heat Q between gas and the environment, greatest first.



30

Special cases of the 1st Law

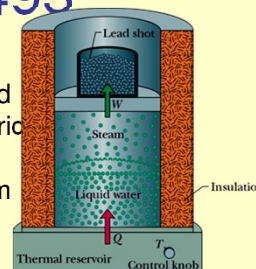
- **Adiabatic Process:** $Q = 0$, $\Delta E_{\text{int}} = -W$ i.e. if the system does work (W is +ve) the change in internal energy is negative.
- **Isovolumetric Process:** $W = 0$, $\Delta E_{\text{int}} = Q$ i.e. $\Delta V = 0$, heat is absorbed or lost.
- **Cyclical Process:** $W = Q$, $\Delta E_{\text{int}} = 0$ i.e. the system is restored to the initial state i , so no change temperature and internal energy is unchanged.
- **Free expansions:** $W = 0$, $Q = 0$, $\Delta E_{\text{int}} = 0$ i.e. adiabatic process where no heat is transferred and no work is done on or by the system.



31

Sample Problem p493

- Let 1.00 kg of liquid water be at 100 °C be converted to steam at 100 °C by boiling at standard atmospheric pressure ($1.00 \text{ atm} = 1.01 \times 10^5 \text{ Pa}$) in an arrangement shown. The volume of the water changes from $1.00 \times 10^{-3} \text{ m}^3$ as a liquid to 1.671 m^3 as steam.
 - How much work is done by the system during this process?
 - How much energy is transferred as heat during the process?
 - What is the change in the system's internal energy?



32

Heat Transfer Mechanisms

- Thermal conduction
 - Metal spoon left in the pot when cooking
- Convection
 - Heater in the room
- Radiation
 - Solar energy

